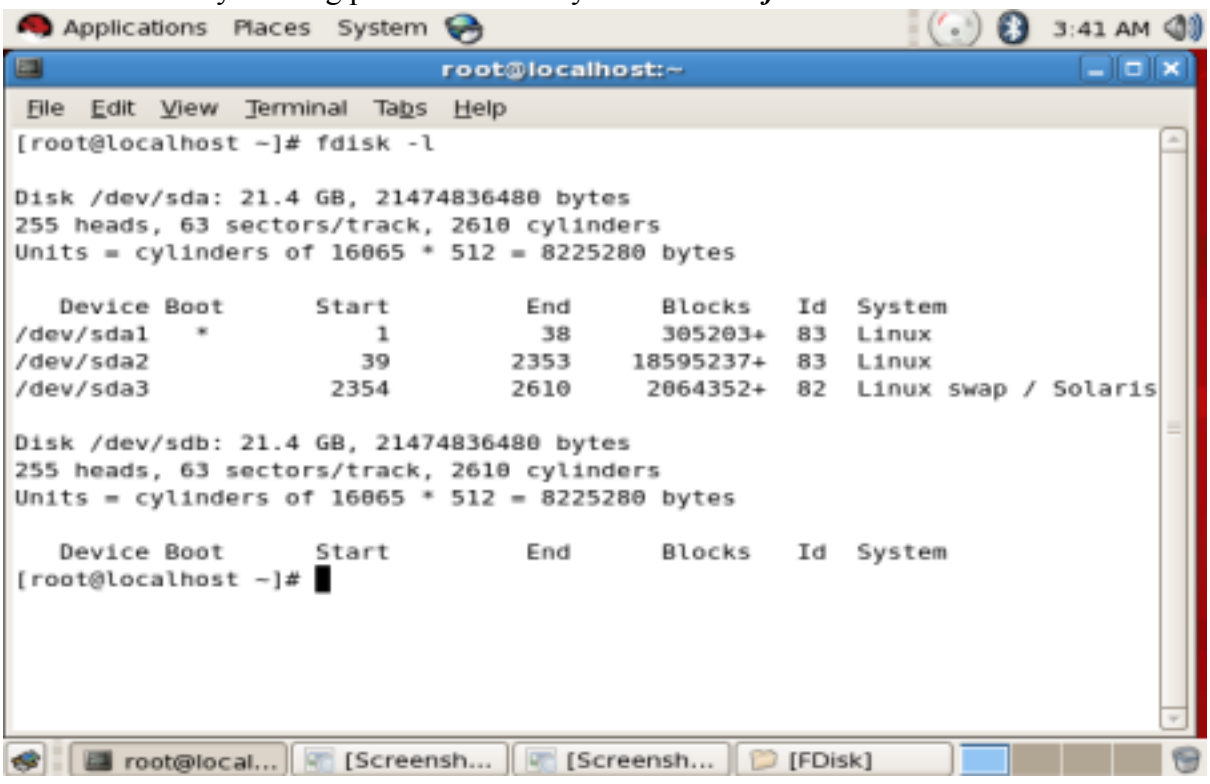
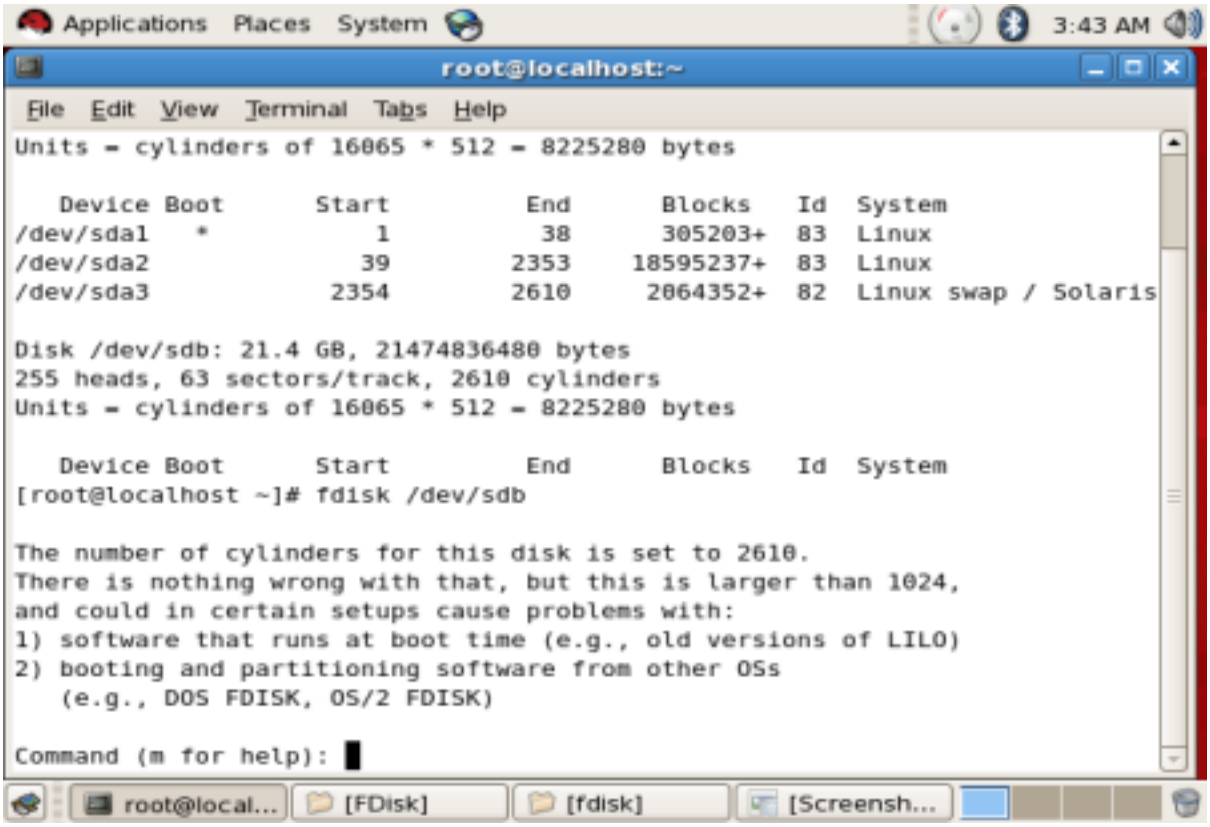
Aim: Demonstrate Disk Management commands for disk partitioning (create & delete partitions) in LINUX (Red Hat Enterprise Linux).

Theory : We know that Linux is a "multi-user" and "multi-tasking" system. We need to be in the "**root**" profile in order to **create/manage/delete** disk partitions. Enter RHEL through the "**root**" login in order to create partitions in your system. After you are in through "**root**" provide your "**username"** and "**password"** and log-in into the system:

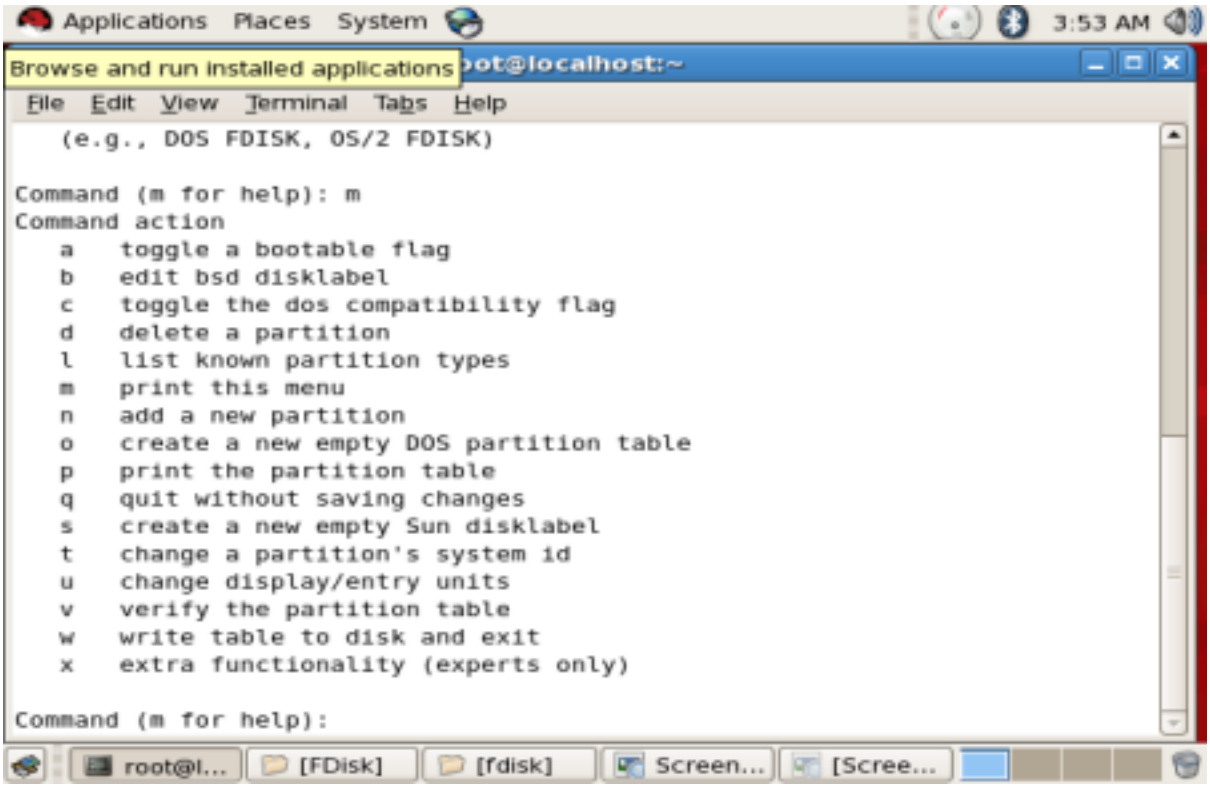
To **list** the already existing partitions in the system we use "***fdisk -l***" command:



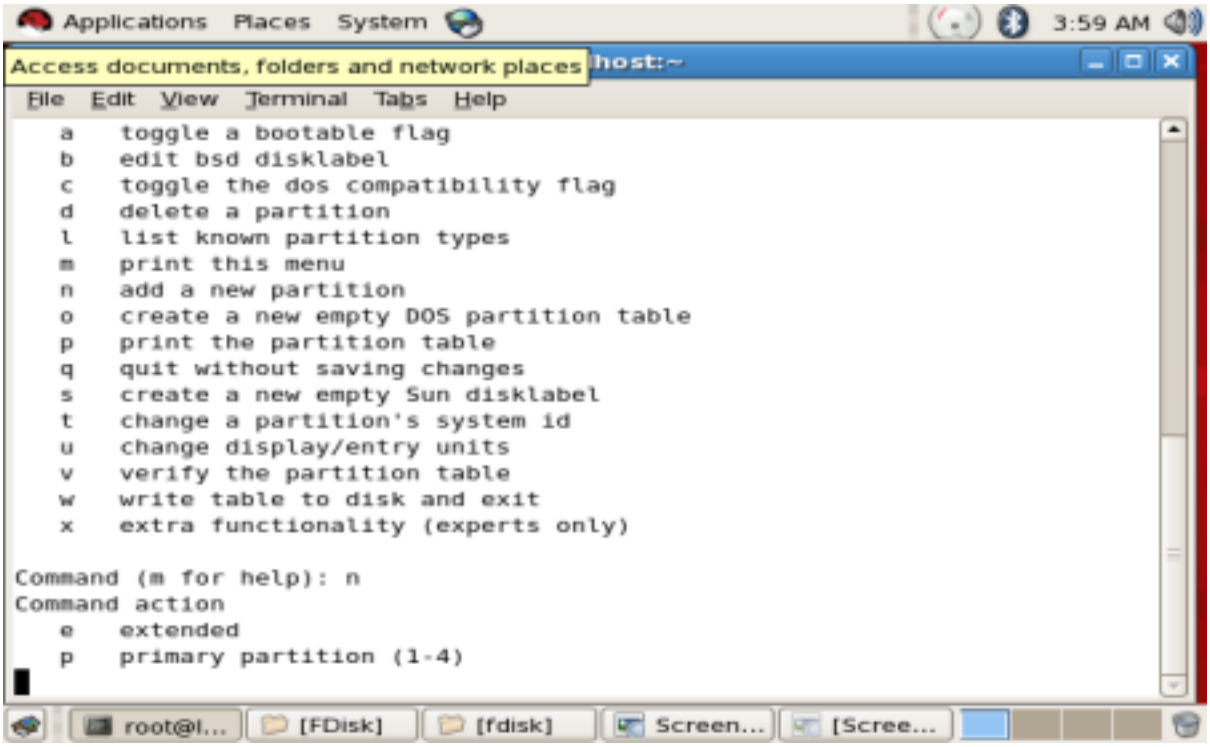
To get into the new storage use "***fdisk /dev/sdb***" which can be seen during **booting** process:



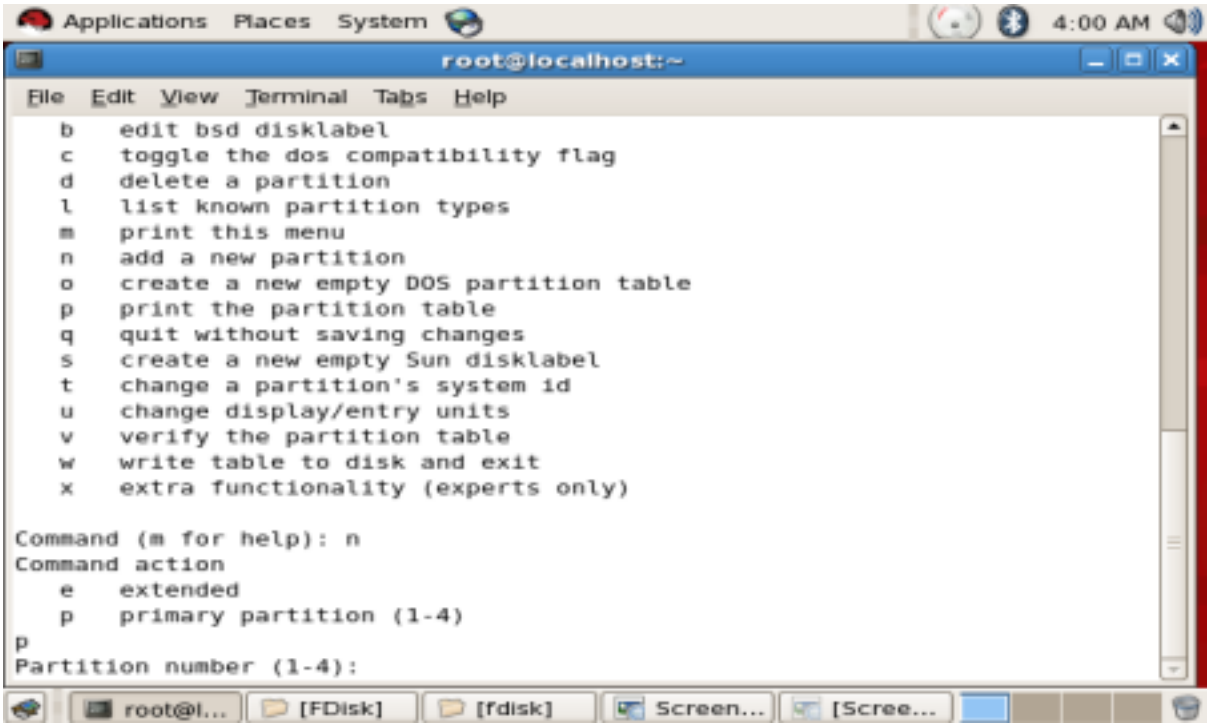
In order to know all the commands associated with disk, you can use the "***m***" command which displays all the associated commands in "*fdisk*".



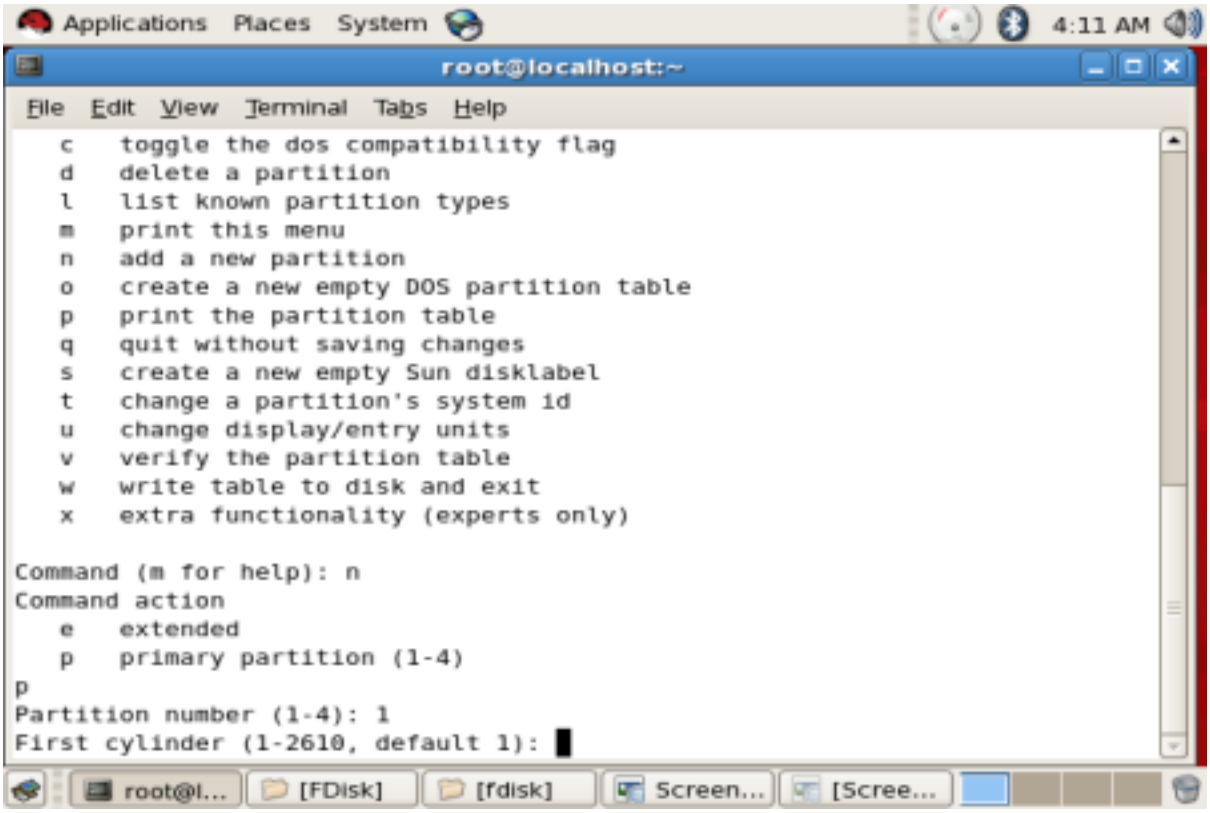
We have checked that we use "***-l***" in order to check for the known partition types. Now, in order to create a new partition we use "***n***".



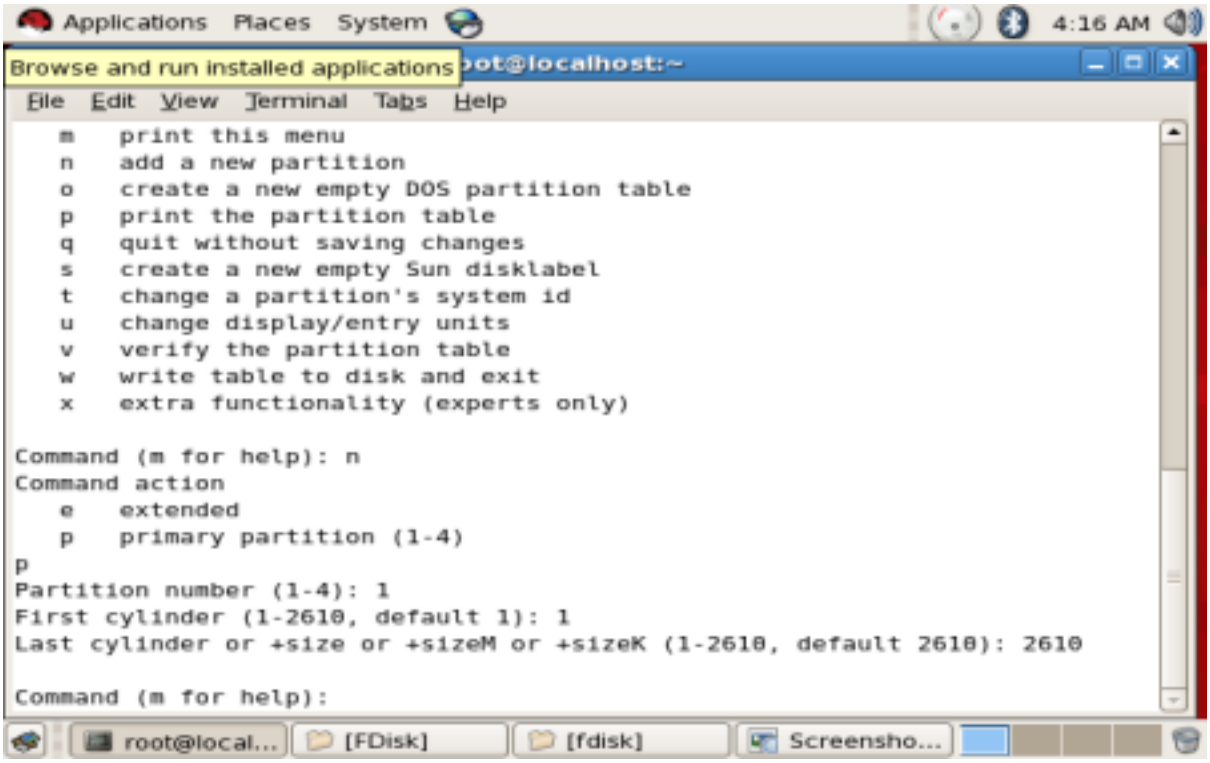
After using the "***n***" for new partition we use "***p***" in order to create a "**primary partition"**.



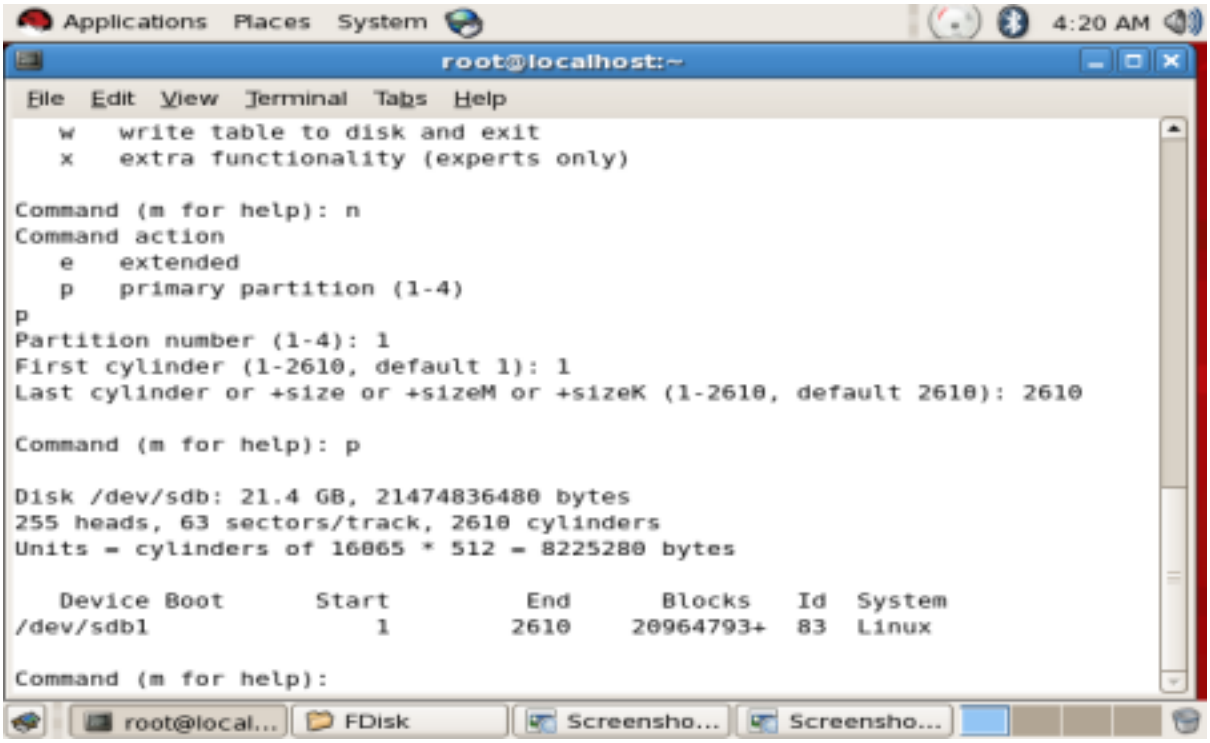
We can use the default section values, 1/2/3/4 to decide on the partition number which will be used by us. We will choose "1" generically, any other number can also be chosen.



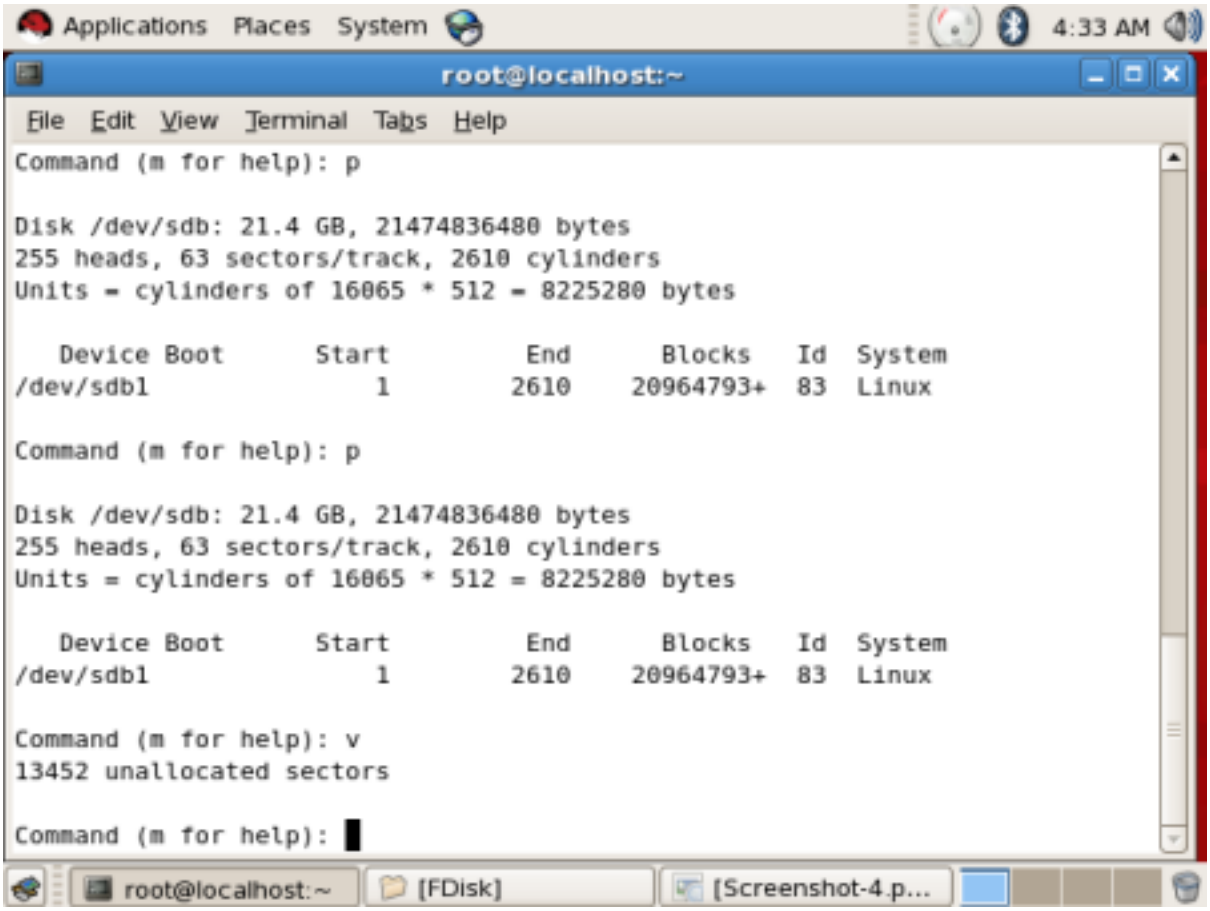
We are allocating the **first cylinder at partition 1** and also mentioning the default position at last, 2610 so that "**Partition space 1**" can be created. Generically while creating partitions we have to look at the prompts for space being allocated to us and specify it accordingly.



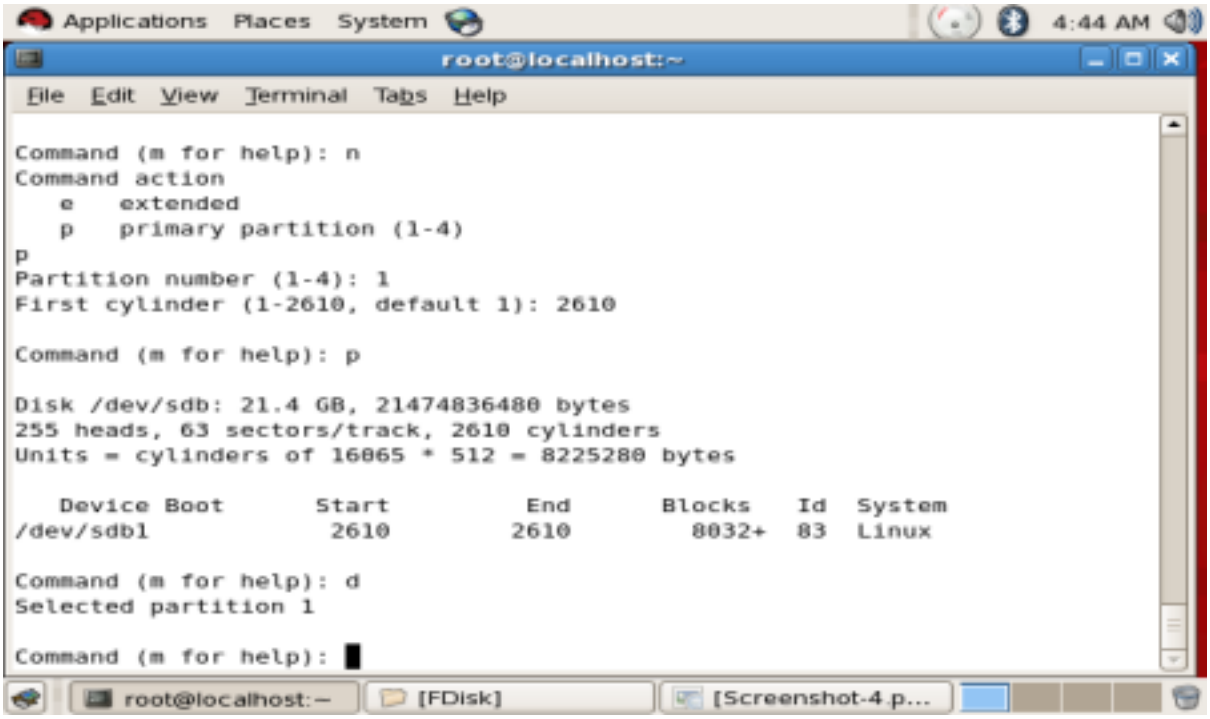
After creating the 1st partition according to commands in disk partitioning we use "***p***" to print the details of the partition created:



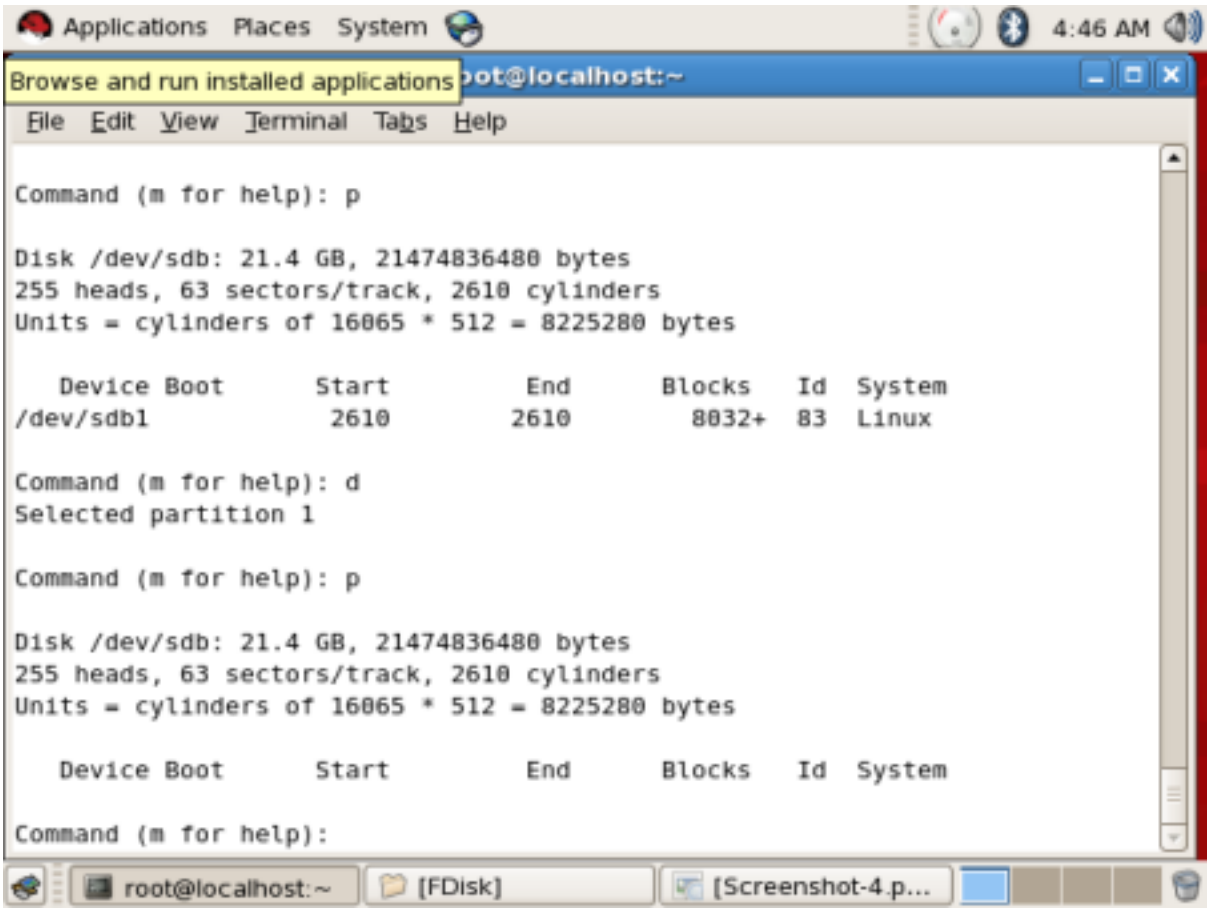
We use "***v***" to verify the unallocated sectors if we create further partition tables:



In order to **delete the partitions** we use the "***d***" command whose output is as follows:



In order to check if the partition has been deleted we use the "***p***" command which is used to print all the partitions which exist in the system.



Aim: Build a shell script in order to show the various system configuration information.

Theory: **Shell Variables**

The variable is a place holder for storing the data. The value of the variable can be changed during the program execution. The value assigned could be a number, text, filename, device, or any other type of data. A variable is nothing more than a pointer to the actual data. The shell enables you to create, assign, and delete variables. Shell variables can be used at the command line and/or used within shell programs. While similar to other programming language variables, shell variables do have some different characteristics such as:

- shell variables do not need to be declared
- all shell variables are of type string

There are two types of variables:

- User defined variables (UDV)
- System Defined variables (SDV)

How to assign value to a variable?

Syntax: variable-name=value 'value' is assigned to given 'variable name' and Value must be onright side = sign.

System variables

Created and maintained by Linux/Unix itself. This type of variable is defined in CAPITAL LETTERS. **These variables are generally set using the "export" command**. Some of the commonly required variables are as follows.

PS1 //Command prompt
HOME //your home directory

PWD //your present working directory

LOGNAME //your login name

USER // same as login name

HISTSIZE // no. of history commands to be stored in history file

HOSTNAME //your host name

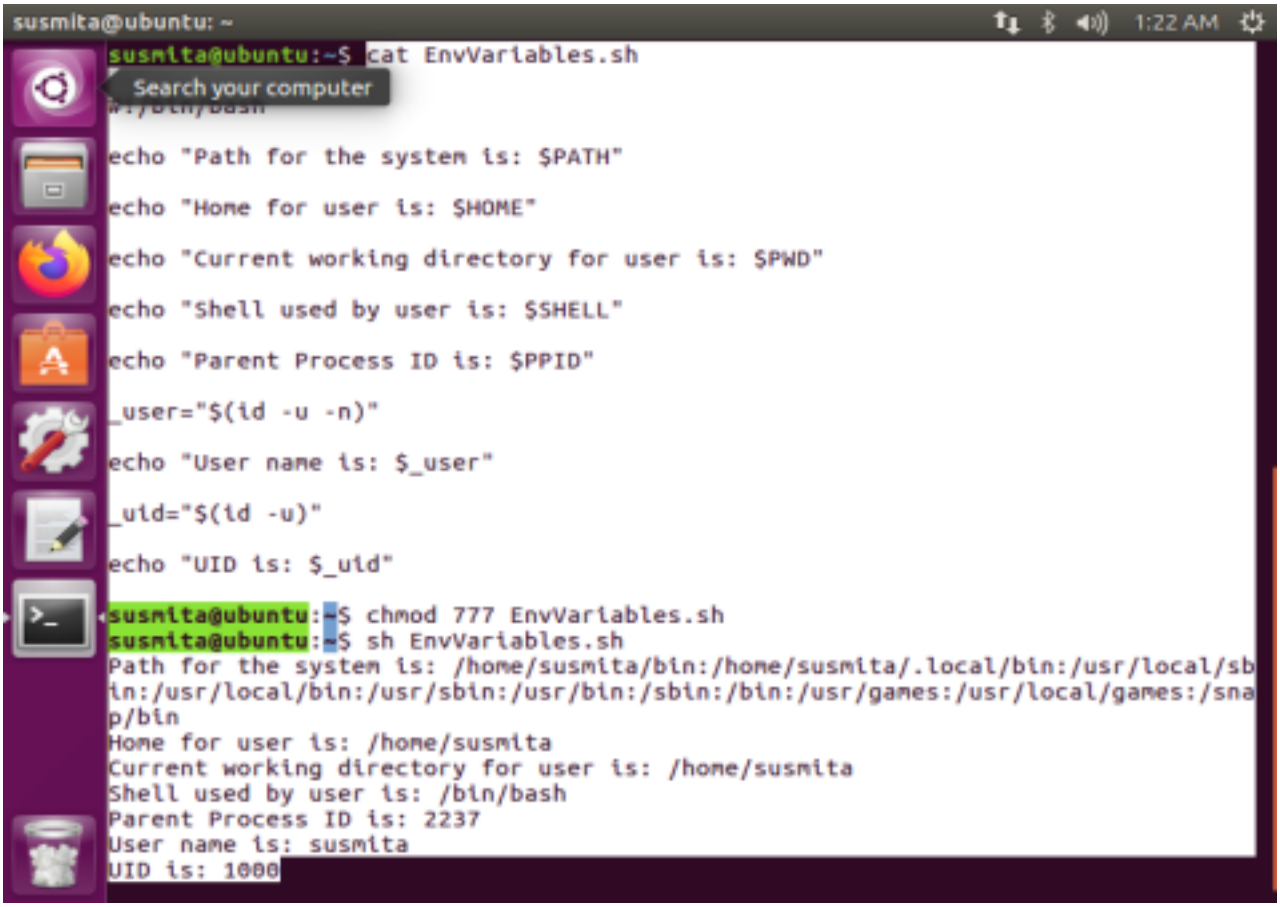
SHELL //your login shell

PPID //Process id of parent process

For system allocated variables like username/ UID etc. we cannot directly access them like the environment variables studied earlier as these variables are created by the system on the user's behalf. Irrespective of system defined or user defined variables, they are retrieved using the $ symbol.

Following is the demonstration for a shell script with the problem statement to **show the**

various system configuration information.



Aim: Build a shell script that accepts the hostname and IP address as command-line arguments and adds them to the "/etc/hosts" file.

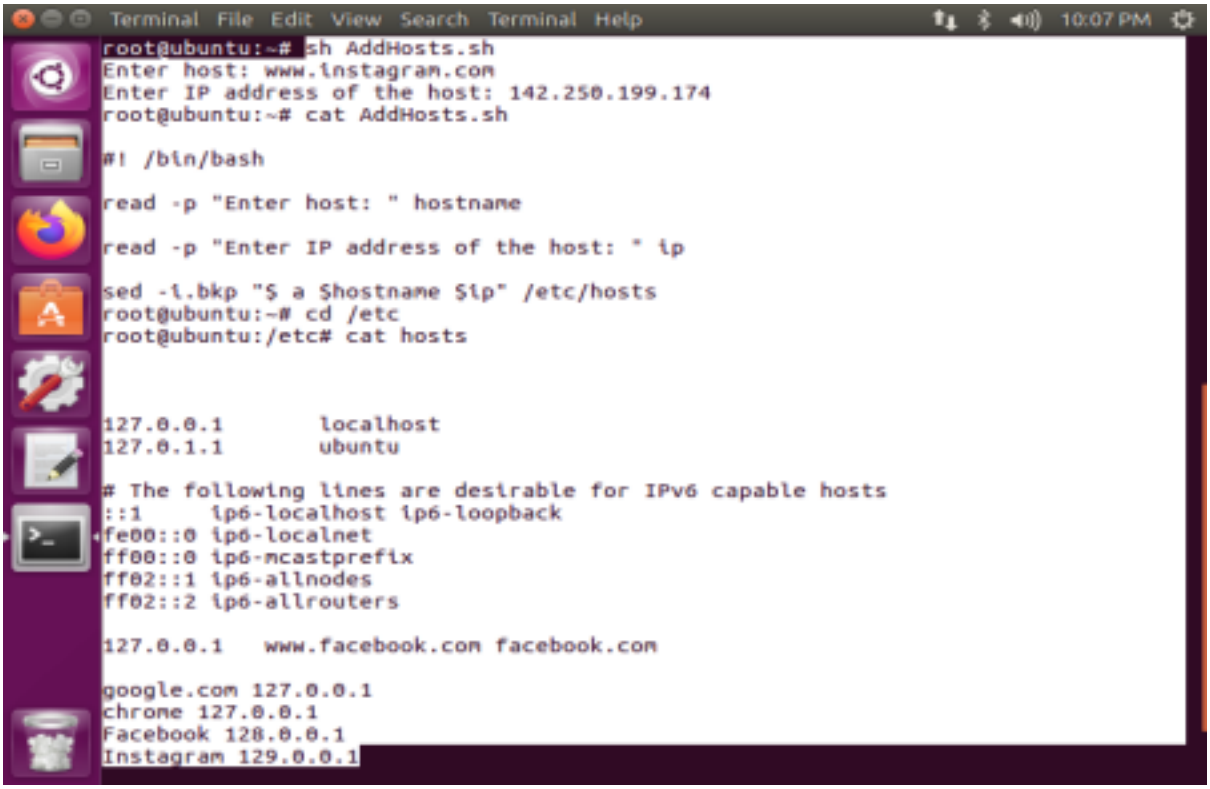
Theory: The "hosts" file is used to map domain names (hostnames) to IP addresses. It is a plain text file used by all operating systems including, Linux, Windows, and macOS. The hosts file has priority over DNS (Domain Name Server). When you type in the domain name of a web site you want to visit, the domain name must be translated into its corresponding IP Address. The operating system first checks its "hosts" file for the corresponding domain, and if there is no entry for the domain, it will query the configured DNS servers to resolve the specified domain name. This affects only the computer on which the change is made, rather than how the domain is resolved worldwide.

Using the "hosts" file to map a domain to an IP address is particularly useful when you want to test your website without changing the domain DNS settings. For example, you are migrating your website to a new server, and you want to verify whether it is fully functional before pointing the domain to the new server. The hosts file can also be used to block websites on your computer.

Within this context of utilities of the "hosts" file we will explore in this experiment how we can edit the "hosts" file using a shell script. The underlying objective is to understand how to add hosts to our system, and also how to ensure the safety & security of our Linux system by filtering only those

hosts/ websites which are authenticated by the particular administrator of the system. This measure can be further extended to different types of users, groups as discussed in earlier experiments.

In this program we check the utility of shell scripts w.r.t **to adding the hosts directly in our Linux system** by using a shell script.



Aim: Assess remote network applications viz. “Remote server access/administration” with SSH (Secure Shell) & PuTTY (where TTY stands for terminal teletype).

Theory: PuTTY is a free, light-weight, open-source terminal emulator. It is a serial port network file transfer application. It supports several network protocols, including SCP, SSH, Telnet, rlogin, and raw socket connection. PuTTY is an SSH and telnet client, developed originally by Simon Tatham for the Windows platform; (also available for installation on Linux platforms).

An SSH client is a software program which uses the secure shell protocol to connect to a remote computer. You will find PuTTY useful if you want to access an account on a Unix or other multi user system from a PC (for example your own or one in an internet cafe). PuTTY is an alternative to telnet clients. Its primary advantage is that SSH provides a secure, encrypted connection to the remote system. It's also small and self-contained and can be carried around on a floppy disk.

Use of PuTTY: In a multi-user operating system like Unix, the interface is generally of command line type, just like the command prompt or MS-DOS. As such the user needs to type in the command in the command line program to get anything processed by the system. Generally, these commands can quickly be run over a network from a different computer on a different location (client) and the response is transferred over the network to the client.

The arrangement mentioned above is made possible with the help of network protocols like SSH, Telnet, Rlogin, etc. Interestingly, users can give commands to multiple computers simultaneously.

SSH (Secure Shell) protocol is a cryptographic network protocol that allows you to access an internet server while encrypting any information sent to that server. Some of the other protocols include Telnet, Rlogin only if either you are connected to a Unix system or you have a login account on a web server (a shell account). PuTTY is one such application that enables this kind of transfer.

How to Manage a Session in PuTTY

It is the preliminary panel where you get to specify specific options to open a session.

- The **Host Name (or IP Address)** bar is where a user will input the name or IP address of the server they want to connect.
- **Connection type** of radio buttons allows users to choose from the kind of network they are planning to connect.
- The Port bar is the section that is filled automatically on selecting the type of connection. However, if you choose the **Raw** type, the bar stays blank and requires the user to enter the **port** manually.
- Upon selecting Serial as the connection type, the **Host Name and Port bars** will be replaced by **Serial Line and Speed**.

The "**Load, save or delete a stored session"** section is to set some connection setting without having to type all the details again when needed.

- Once you save it, it can just select on the saved session and click on Load. The saved settings will appear on their respective boxes in the configuration panel.
- The panel permits to modify a saved session by first loading a session, editing everything you want to change and then clicking on "save" button.
- Users have the option to delete a session as well.

The **Close Window On Exit** option helps in deciding whether the PuTTY terminal will close as soon as the session ends or restarts the session on the termination.

What is Logging in PuTTY?

This configuration panel saved the log files of your PuTTY sessions which can be used for debugging and analysis purposes. Users can choose the type of data you want to log in this window.

What is Terminal in PuTTY?

The section has a variety of options to decide how the texts in the window should appear. Whether you want the text to come in the next line as soon as it reaches the right edge of the window or you want to interpret the cursor position.

Configure Keyboard setting in PuTTY.

With the option, users can modify the behavior of 'backspace,' 'home' and 'end' keys, and several other keys to coordinate with the server settings.

What is Bell in PuTTY?

It lets PuTTY make an alert sound as and when you want it to function.

Manage Data in PuTTY.

The auto-login option dismisses the need to type the username every time. It can also specify the terminal needs using this panel.

A proxy setting in PuTTY

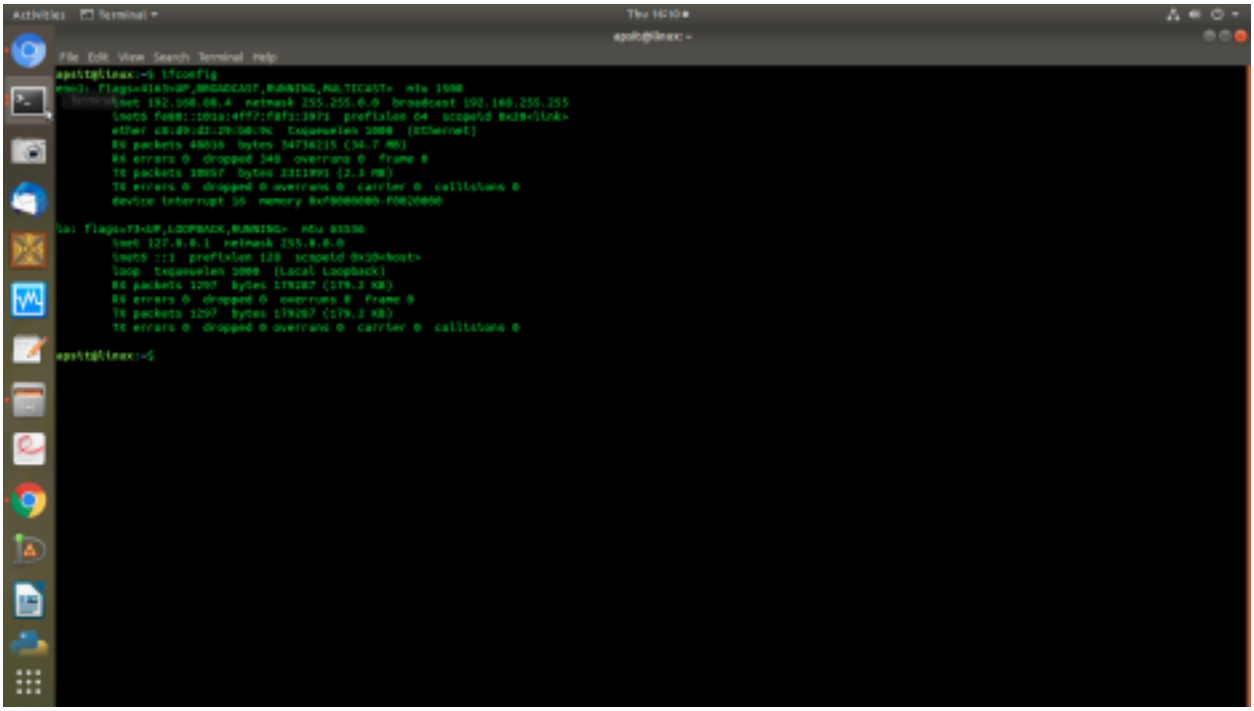
Permits configuration of various types of proxies used while making the network connections.

What are Telnet, Rlogin, and SSH in PuTTY?

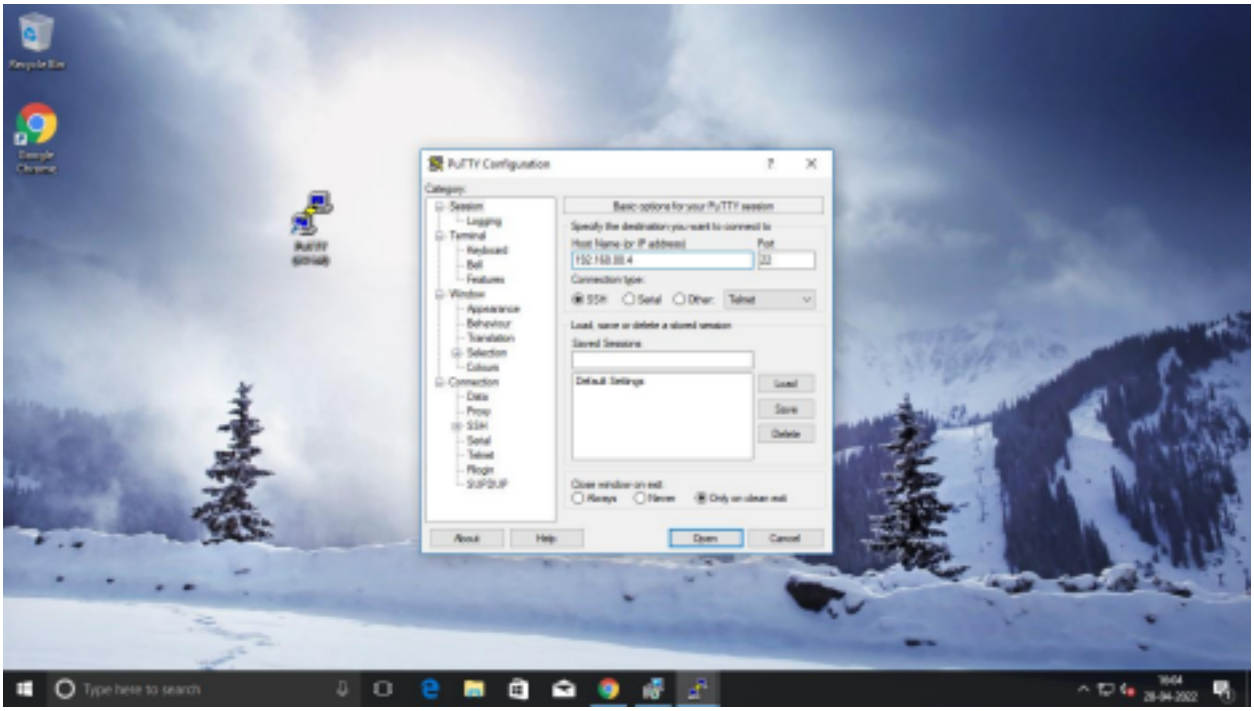
These panels allow configuration of session-related options like changing the mode for negotiation between the server and client, allowing an automated form of login on the server, etc.

Accessing remote networks using SSH & Putty:

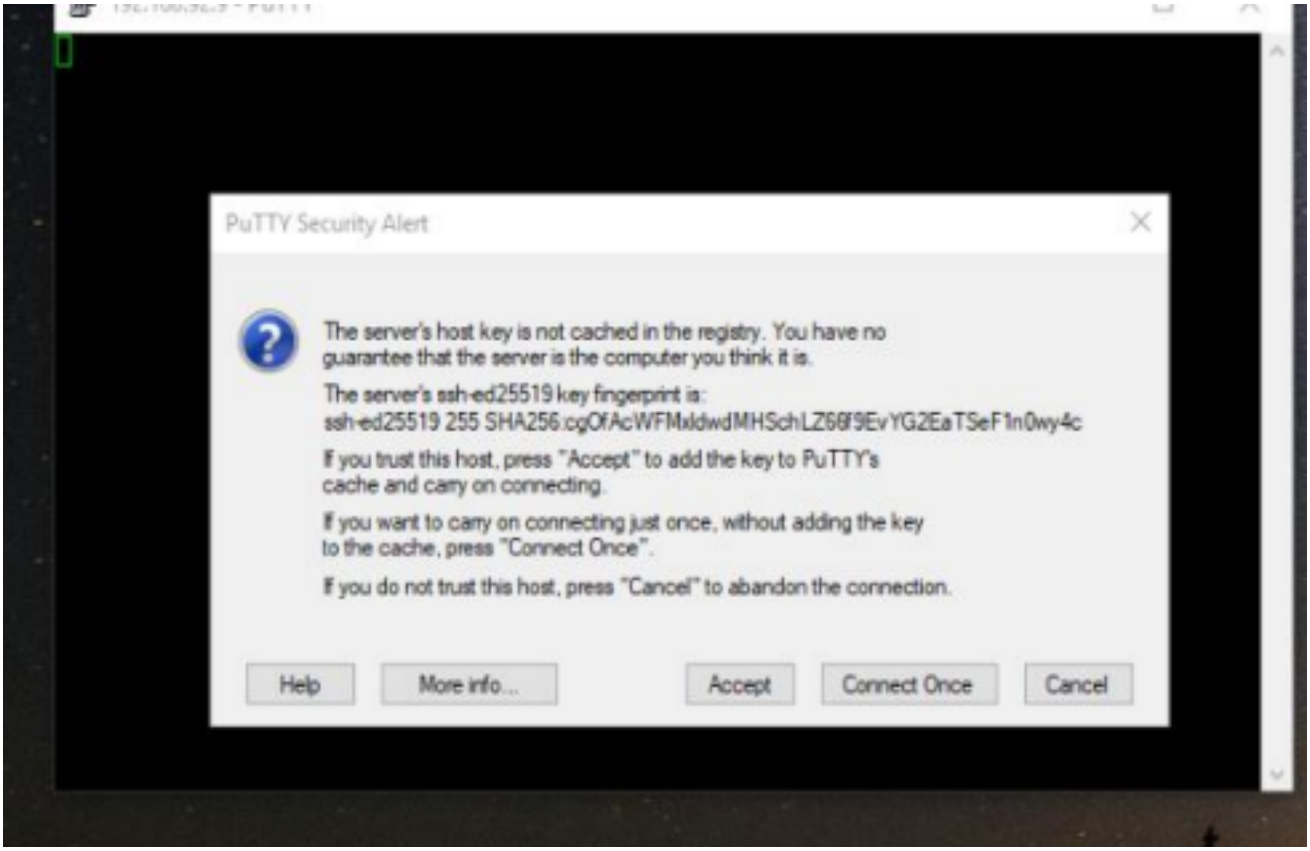
The following terminal with the **IP address (192.168.88.4)** is our Server machine which we try to connect to through the **PuTTY** (installed Windows/Linux enabled terminal). We obtain this IP using the command **ifconfig** in our Linux system.



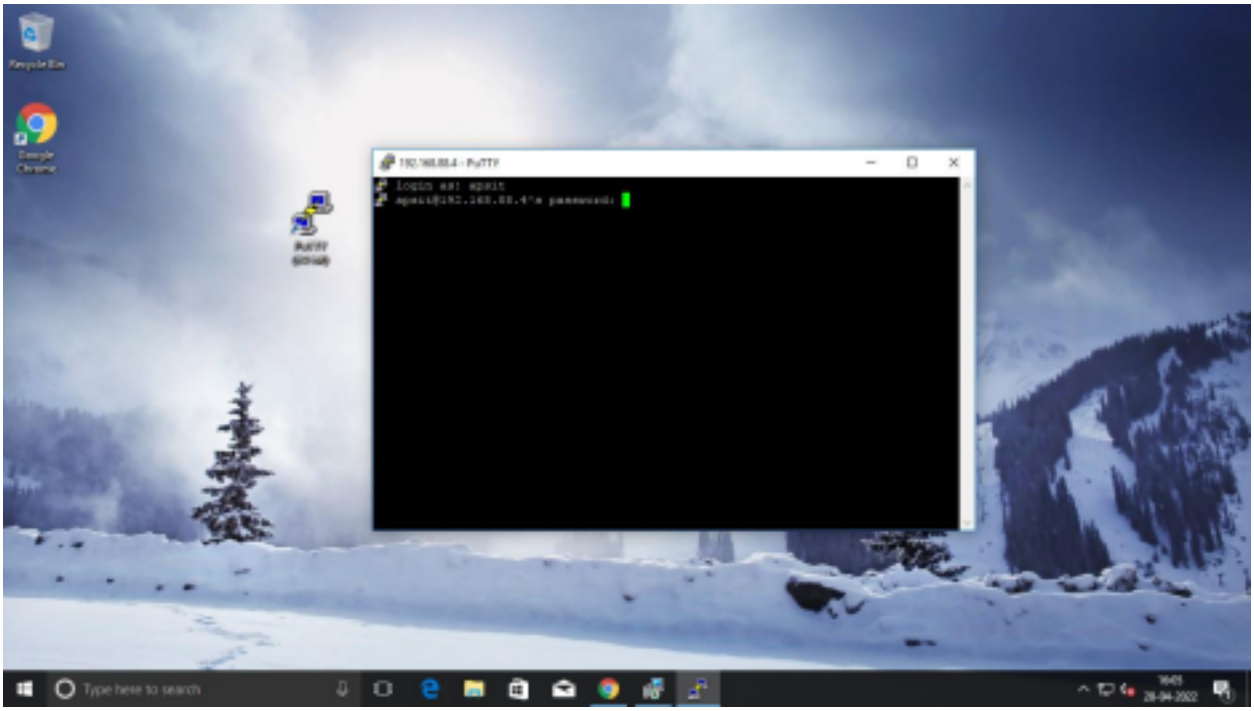
We will download the Putty application on our Windows Client machine. Post this, in order to connect to the "Server" we login through the "**Putty Configuration**" window. On terminal (Client side) we will provide the IP address of the machine we want to connect to as seen.



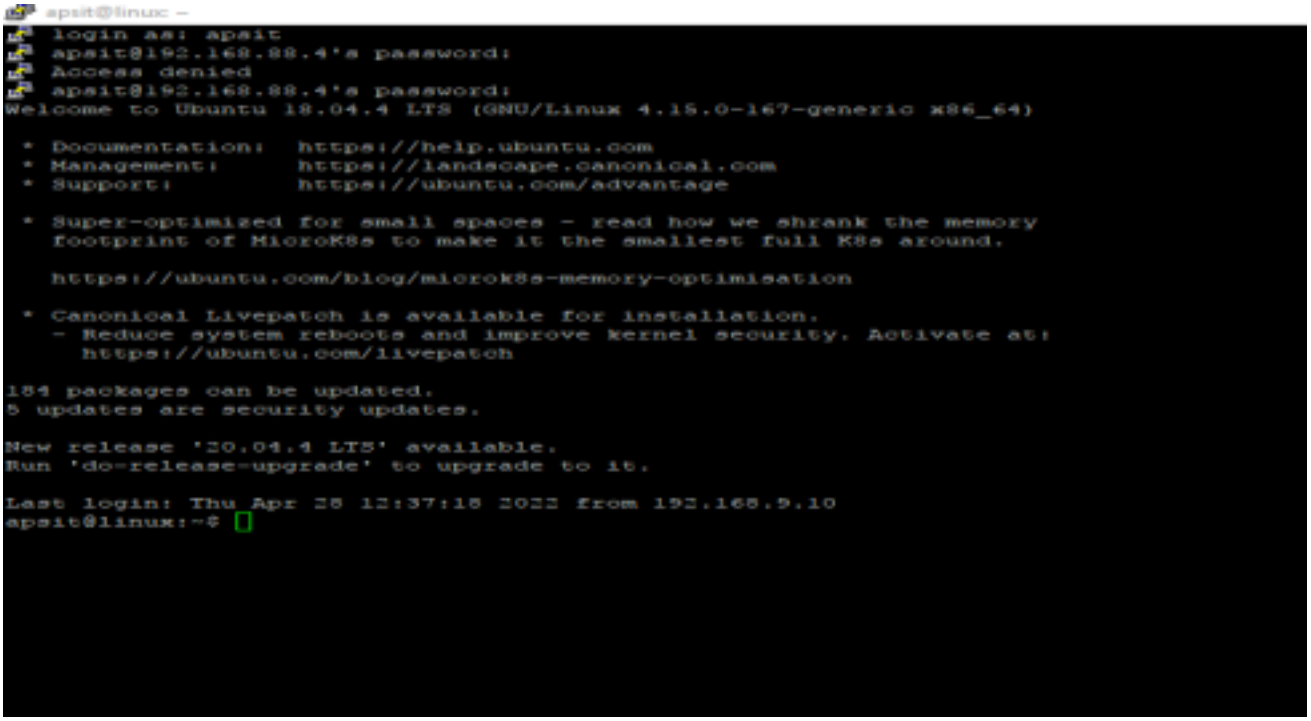
In the following screen, we can observe that the "**Putty Security Alert**" screen prompts the user to enter choices between: **"Accept", "Connect Once", "Cancel".** We will choose **the "Accept"** option so that we can make a trusted connection with the **Server.**



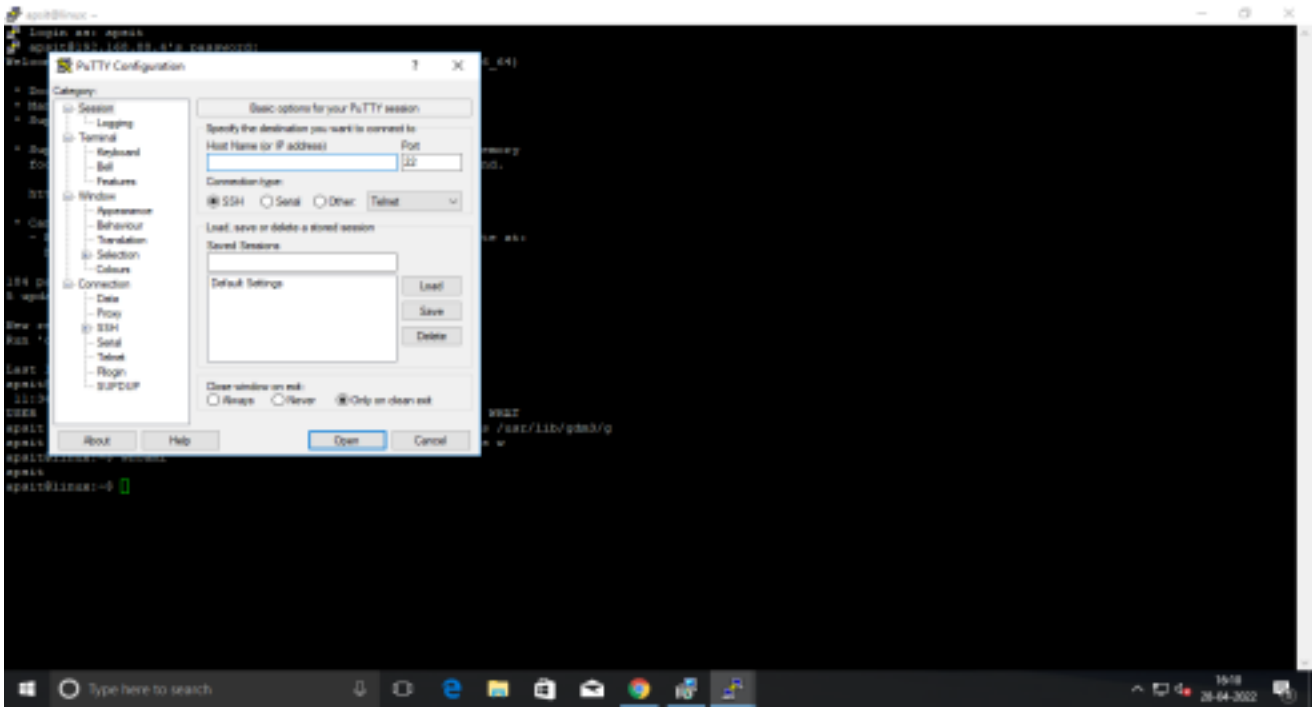
Login with the credentials of "**login as**" and "**password**" of the system for PuTTY:



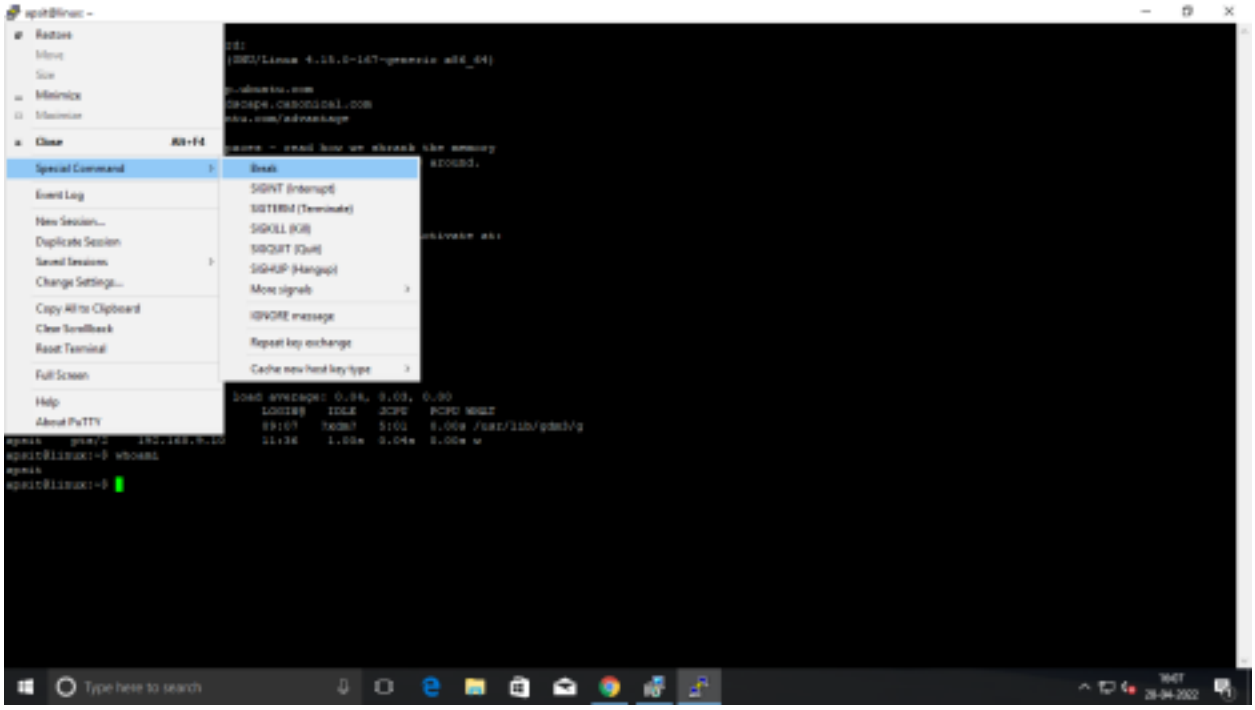
Once we login the immediate screen visible is informative with details like the release version of Ubuntu, the no. of packages updated and also others like Documentation, Management & Support info.



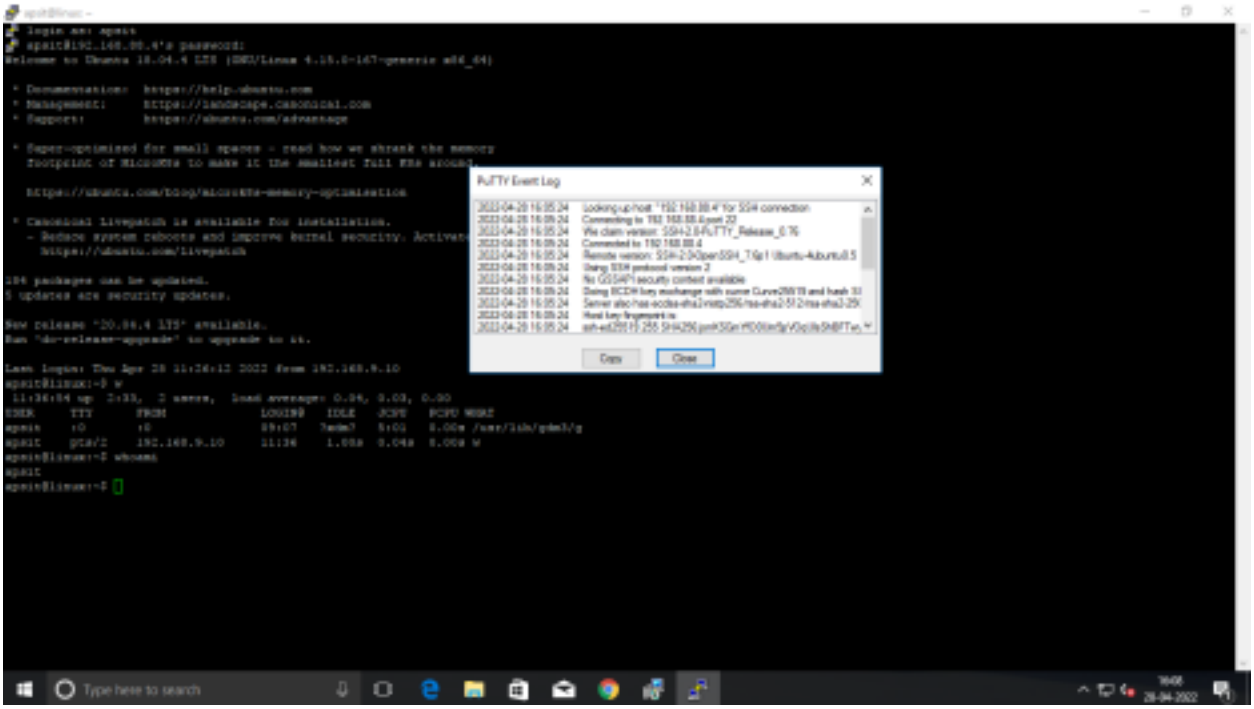
Whenever you create a "**New connection**" use the IP address of the machine you must connect to:



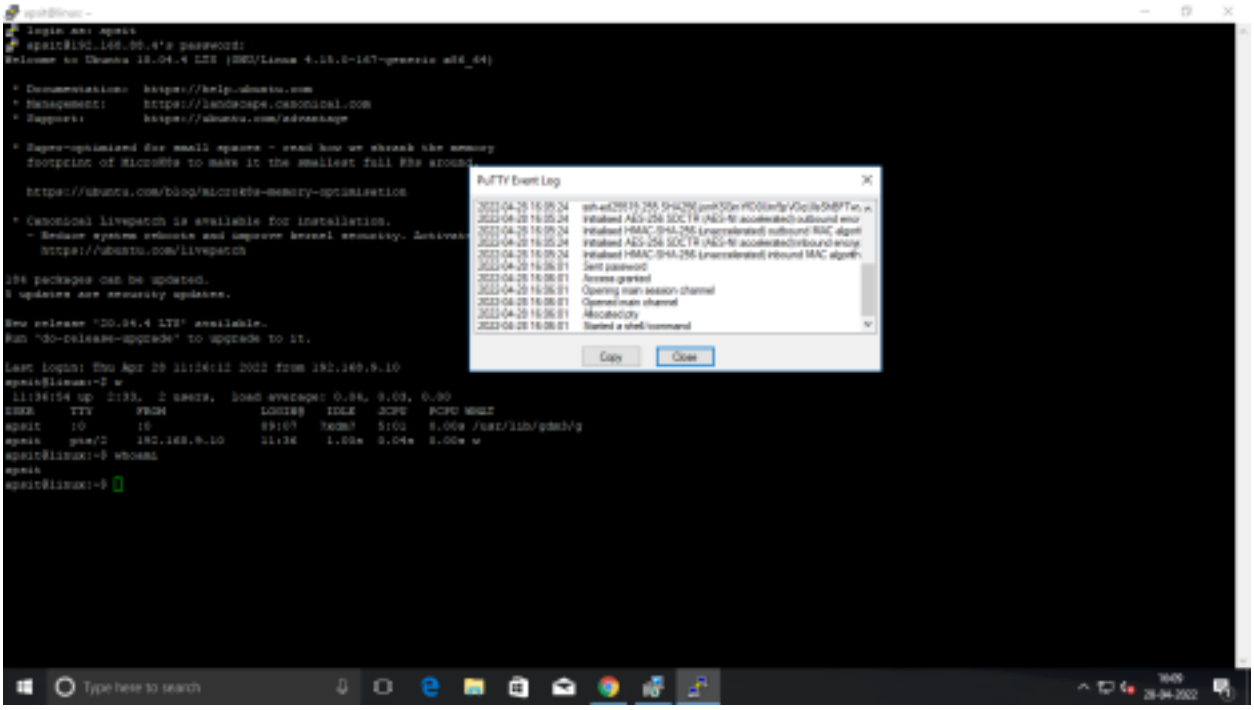
Special commands available (on Linux system) and that can be also be accessed through **remote login via PuTTY are SIGINT/SIGTERM/SIGKILL** to mention a few:



Software interrupts on Linux and Unix systems are made via signals. **There are many different Linux signals, but a few stand out and are important to understand and know: SIGINT, SIGTERM, and SIGKILL.** The basic Linux signals all have a number (1-30+). After a little while, a proficient Linux user will generally know one or more of these. For example, the SIGTERM signal matches with number 15, and signal 9 (SIGKILL) is likely the most known one as it allows one to forcefully terminate a process, a SIGTERM signal is used to send a warning signal to a process; however, the process/program may choose to ignore such a signal.



Th e “**PuTTY Event Log**” maintains a log of all the activities being executed on the machine.



For a different session with Server IP (192.168.7.43) we observe in the following screens that the same output is observable when we use the command “w” on the **Server and on Client machine**:

Aim: Configuration of NFS (Network File System) Server and transfer files to a Linux Client.

Theory: NFS (Network File System) is a distributed file system. An NFS server has one or more file systems that are mounted by NFS clients. These file systems are mounted by using the standard UNIX mount commands. NFS is the

standard method to share files between Linux and Unix-like systems.

NFS is a networking protocol for distributed file sharing. A file system defines the way files are stored and retrieved from storage devices, such as hard disk drives, solid-state drives and on tape drives across networks. It enables network administrators to share all or a portion of a file system on a networked server to make it accessible to remote computer users. NFS is one of the most widely used protocols for file servers. Cloud vendors also implement the NFS protocol for cloud storage, including Amazon Elastic File System, NFS file shares in Microsoft Azure and Google Cloud Filestore.

How does the Network File System work?

NFS is a client-server protocol. An NFS Server is a host that meets the following requirements:

- has NFS installed
- has at least one network connection for sharing NFS resources; and
- is configured to accept and respond to NFS requests over the network connection. An NFS Client is a host

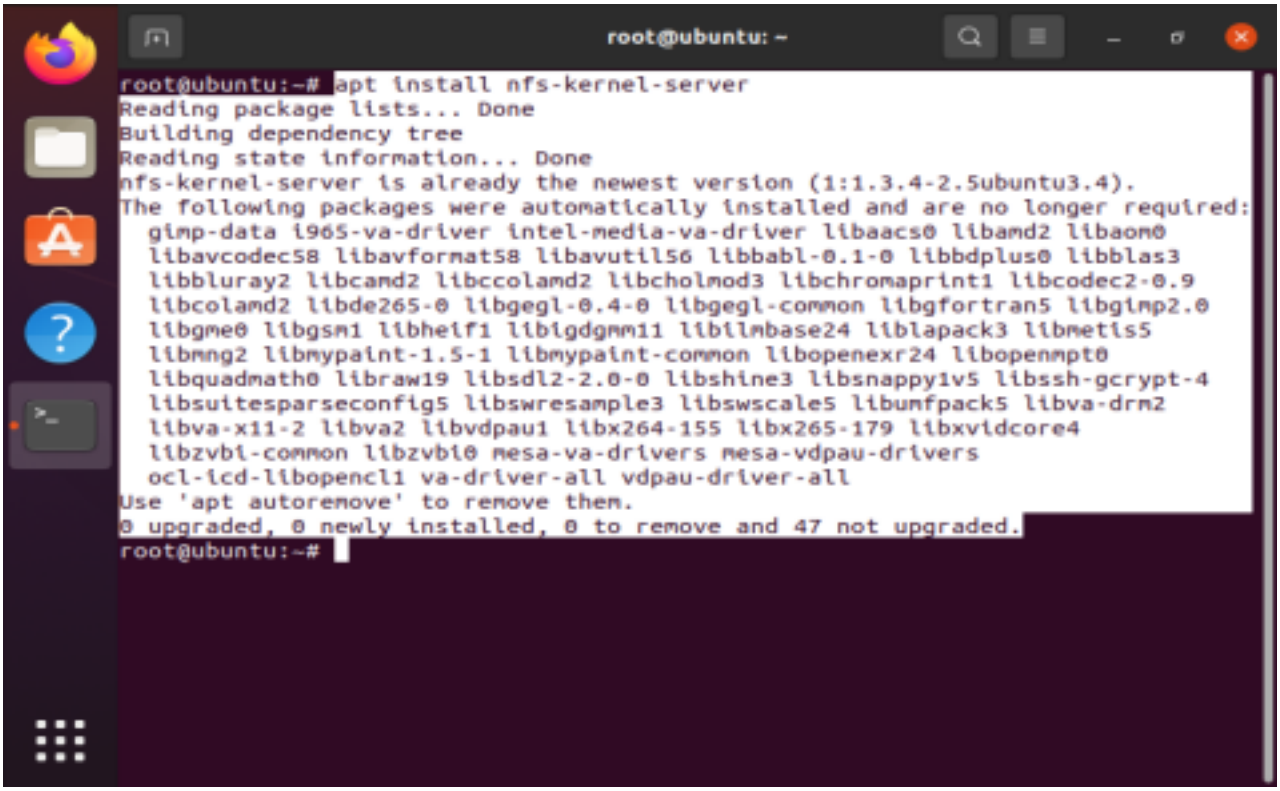
that meets the following requirements:

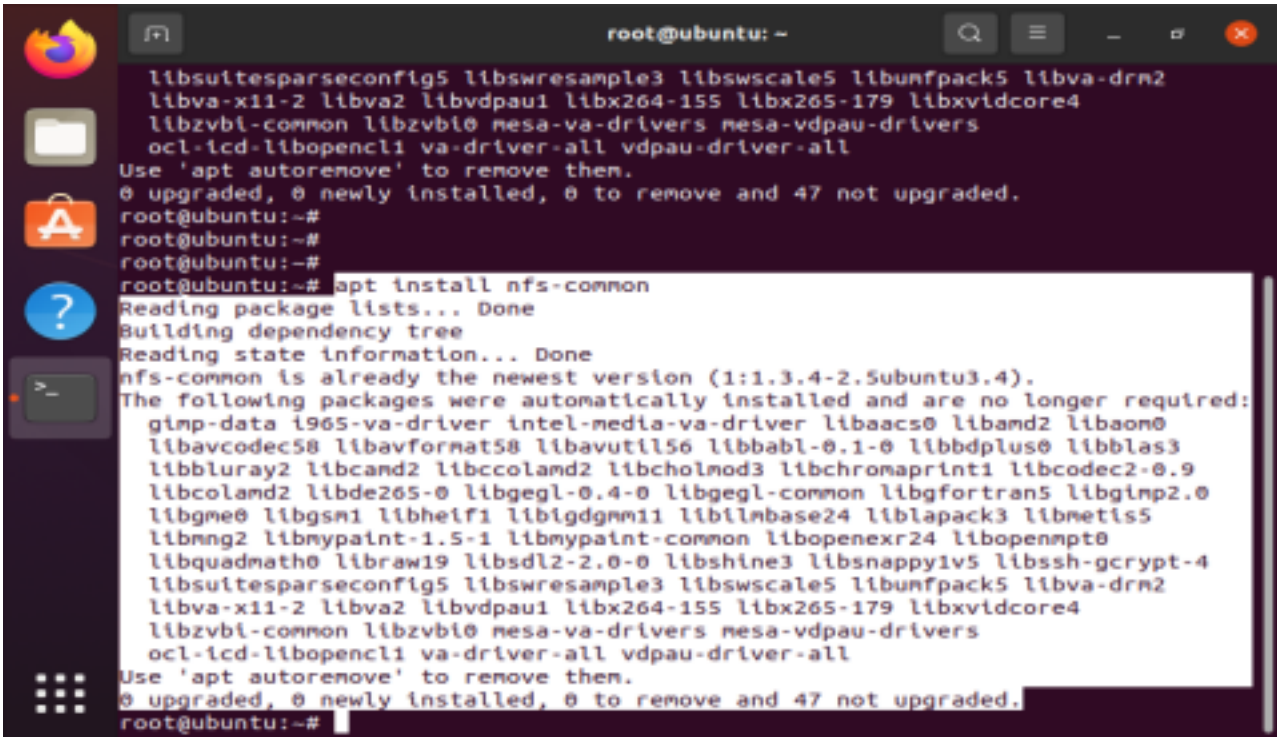
- has NFS client software installed
- has network connectivity to an NFS server
- is authorized to access resources on the NFS server; and
- is configured to send and receive NFS requests over the network connection.

NFS was initially conceived as a method for sharing file systems across workgroups using UNIX. It is still often used for ad hoc sharing of resources.

Installation of NFS Server

Install the package for "**nfs-kernel-server**" on your machine.





Create the folder that needs to be shared,

```
root@ubuntu:~#
root@ubuntu:~# cd /mnt
root@ubuntu:/mnt#
root@ubuntu:/mnt#
root@ubuntu:/mnt# ls -lrt
total 0
root@ubuntu:/mnt#
root@ubuntu:/mnt# mkdir nfsshare
root@ubuntu:/mnt#
root@ubuntu:/mnt# ls -lrt
total 4
drwxr-xr-x 2 root root 4096 Apr 27 07:35 nfsshare
root@ubuntu:/mnt#
```

Provide permissions as below,

```
root@ubuntu:/mnt#
root@ubuntu:/mnt# chown -R nobody:nogroup nfsshare
root@ubuntu:/mnt#
root@ubuntu:/mnt# ls -lrt
total 4
drwxr-xr-x 2 nobody nogroup 4096 Apr 27 07:35 nfsshare
root@ubuntu:/mnt#
root@ubuntu:/mnt#
```

```
root@ubuntu:/mnt#
root@ubuntu:/mnt# chmod -R 777 nfsshare/
root@ubuntu:/mnt#
root@ubuntu:/mnt#
root@ubuntu:/mnt# ls -lrt
total 4
drwxrwxrwx 2 nobody nogroup 4096 Apr 27 07:35 nfsshare
root@ubuntu:/mnt#
root@ubuntu:/mnt#
```

Edit “/etc/exports” with below entry with IP or (*) for all servers,

```
# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes       hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_sub
tree_check)
#
# Example for NFSv4:
# /srv/nfs4        gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes  gss/krb5i(rw,sync,no_subtree_check)

/mnt/nfsshare  192.168.75.129(rw,sync,no_root_squash,insecure)
```

//add the IP address of server with details of the file being shared

Check the status of NFS service after making these changes,

```
root@ubuntu:~#
root@ubuntu:~# systemctl status nfs-kernel-server
● nfs-server.service - NFS server and services
     Loaded: loaded (/lib/systemd/system/nfs-server.service; enabled; vendor p
     Active: active (exited) since Wed 2022-04-27 07:44:41 PDT; 34s ago
    Process: 6243 ExecStartPre=/usr/sbin/exportfs -r (code=exited, status=0/SU
    Process: 6244 ExecStart=/usr/sbin/rpc.nfsd $RPCNFSDARGS (code=exited, stat
   Main PID: 6244 (code=exited, status=0/SUCCESS)

Apr 27 07:44:40 ubuntu systemd[1]: Starting NFS server and services...
Apr 27 07:44:41 ubuntu systemd[1]: Finished NFS server and services.
```

Check the firewall status,

```
root@ubuntu:~# ufw status
Status: inactive
root@ubuntu:~#
root@ubuntu:~#
```

Export the NFS folder and mount the shared folder,

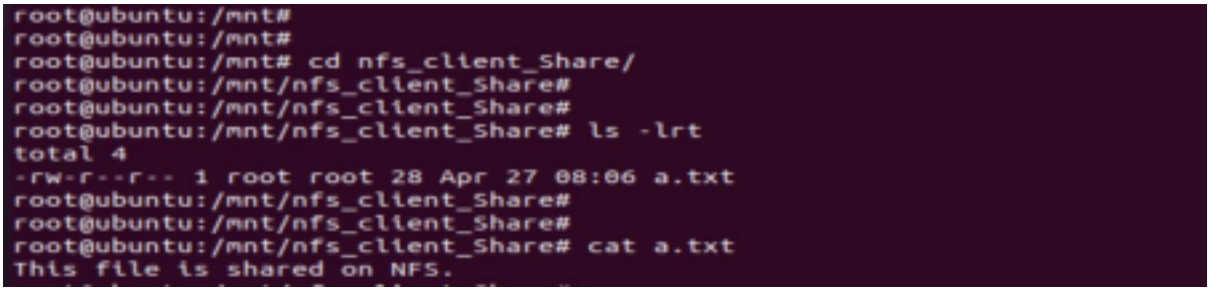
```
root@ubuntu:/mnt#
root@ubuntu:/mnt# exportfs -rav
exportfs: /etc/exports [3]: Neither 'subtree_check' or 'no_subtree_check' speci
fied for export "*:/mnt/nfsshare".
  Assuming default behaviour ('no_subtree_check').
  NOTE: this default has changed since nfs-utils version 1.0.x

exporting *:/mnt/nfsshare
root@ubuntu:/mnt#
root@ubuntu:/mnt#
root@ubuntu:/mnt# mount 192.168.75.129:/mnt/nfsshare  /mnt/nfs_client_Share
root@ubuntu:/mnt#
root@ubuntu:/mnt# ls -lrt
total 8
drwxrwxrwx 2 nobody nogroup 4096 Apr 27 07:35 nfsshare
drwxrwxrwx 2 nobody nogroup 4096 Apr 27 07:35 nfs_client_Share
root@ubuntu:/mnt#
```

Folder acting as server, has a “a.txt” file

```
root@ubuntu:/mnt# cd nfsshare/
root@ubuntu:/mnt/nfsshare#
root@ubuntu:/mnt/nfsshare#
root@ubuntu:/mnt/nfsshare#
root@ubuntu:/mnt/nfsshare#
root@ubuntu:/mnt/nfsshare# ls -lrt
total 4
-rw-r--r-- 1 root root 28 Apr 27 08:06 a.txt
root@ubuntu:/mnt/nfsshare#
root@ubuntu:/mnt/nfsshare#
root@ubuntu:/mnt/nfsshare# cat a.txt
This file is shared on NFS.
root@ubuntu:/mnt/nfsshare#
root@ubuntu:/mnt/nfsshare#
```

Text file acting as a server has a.txt file and is also now visible via Client_share machine


root@ubuntu:/mnt#
root@ubuntu:/mnt#
root@ubuntu:/mnt# cd nfs_client_Share/
root@ubuntu:/mnt/nfs_client_Share#
root@ubuntu:/mnt/nfs_client_Share#
root@ubuntu:/mnt/nfs_client_Share# ls -lrt
total 4
-rw-r--r-- 1 root root 28 Apr 27 08:06 a.txt
root@ubuntu:/mnt/nfs_client_Share#
root@ubuntu:/mnt/nfs_client_Share#
root@ubuntu:/mnt/nfs_client_Share# cat a.txt
This file is shared on NFS.